

HAT-P-3b

R-1282

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-3_210403 · created 2026-07-31T15:13:06+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459307.7958 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1264 +/- 0.005
Transit depth (Rp/Rs)^2	1.6 +/- 0.13 [%]
Orbital Inclination (inc)	89.93 +/- 0.35
Airmass coefficient 1 (a1)	1.0313 +/- 0.006
Airmass coefficient 2 (a2)	-0.028 +/- 0.011
Scatter in the residuals of the lightcurve fit is	0.58 %
Best Comparison Star	#3 - [460, 227]
Optimal Method	PSF photometry
Transit Duration (day)	0.10002 +/- 0.00051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1264 ± 0.005 vs 0.1063 (deviation 3.63σ)
Duration fit vs published	0.10002 d vs 0.0867 d (deviation 23.63σ)
Mid-time O–C	-5.8 min
Depth SNR	22.8
Residual scatter	0.58 %

Light curve

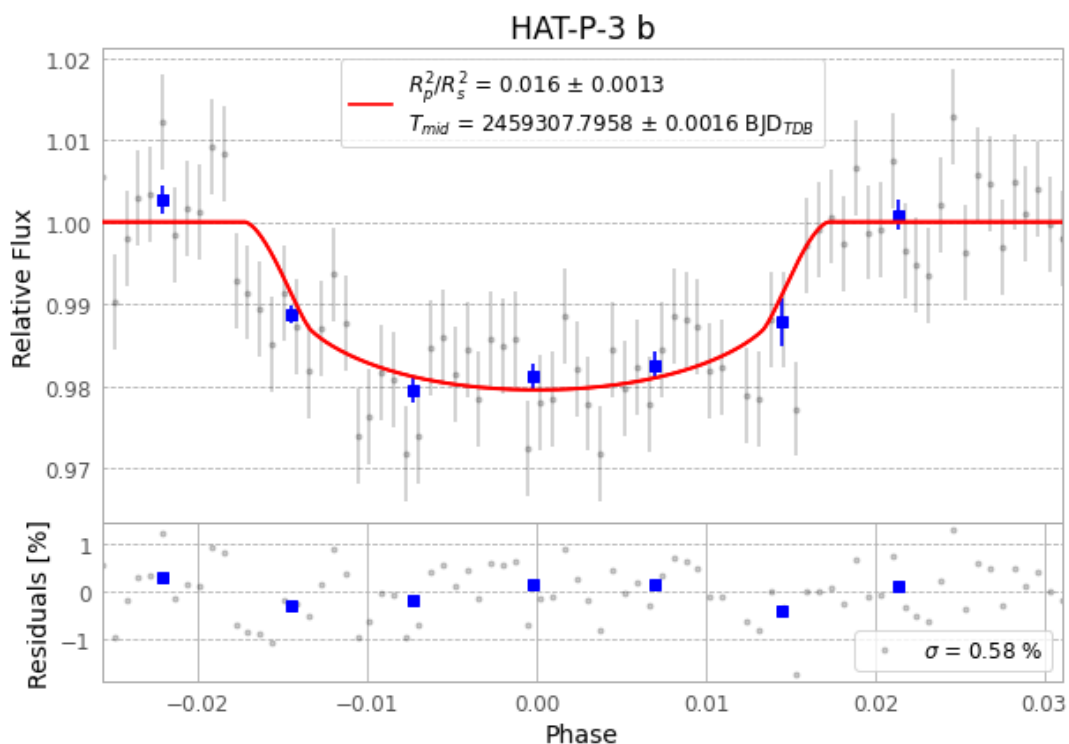


Figure 1 — FinalLightCurve_HAT-P-3 b_2021-04-02.png (SHA-256 8bd93ab229a78fb4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 73b57118d709a297...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

80 FITS frames (53 MB), HATP-3210403051212.FITS → HATP-3210403090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-210403.log (7d62d181be9a...), FinalLightCurve_HAT-P-3 b_2021-04-02.png (8bd93ab229a7...), FinalLightCurve_HAT-P-3 b_2021-04-02.csv (dbdb049e7399...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 15:13Z.